

Números Complejos

$$z = \underbrace{a}_{\text{Real}} + \underbrace{bi}_{\text{Imaginario}} \quad \nabla \text{ Rectangular}$$

Forma Polar

$$i^2 = -1 \quad i = \sqrt{-1}$$

Forma exponencial

$$\operatorname{Re}(z) = a \quad \operatorname{Im}(z) = b \leftarrow \text{solo la } b, \text{ sin la } i$$

$$\text{Si } z = 4 + 3i$$

$$\operatorname{Re}(z) = 4 \quad \operatorname{Im}(z) = 3$$

Suma y resta

$$\text{I) Si } z = 2 - i \text{ y } w = 3 + 2i$$

$$\begin{aligned} z - w \\ (2 - i) - (3 + 2i) \\ 2 - i - 3 - 2i \\ \boxed{-1 - 3i} \end{aligned}$$

$$\begin{aligned} z + w \\ (2 - i) + (3 + 2i) \\ 2 - i + 3 + 2i \\ \boxed{5 + i} \end{aligned}$$

$$\operatorname{Re}(z) = -1$$

$$\operatorname{Im}(z) = -3$$

$$\operatorname{Re}(z) = 5$$

$$\operatorname{Im}(z) = 1$$

Parte de aritmética

§1.2. Operaciones

- Sumar y restar
- Multiplicar
- Conjugado
- Racionalizar
- Dividir
- Factorizar

Multiplicación

$$(a+bi)(c+di)$$

Diagram showing the multiplication of two complex numbers. Red labels 'Re' and 'Im' are above the terms. Blue arcs connect 'a' to 'c' and 'd', and 'b' to 'c' and 'd'. A blue arc also connects 'a' and 'b'.

$$i^2 = -1$$

$$ac + adi + bci + bd(i^2) \rightarrow -1$$

$$\boxed{ac} + \boxed{adi} + \boxed{bci} - \boxed{bd}$$

$$(ac - bd) + (ad + bc)i$$

$$2) (2-i)(3+2i) \quad i^2 = -1$$

$$6 + 4i - 3i - 2i^2$$

$$6 + 4i - 3i + 2$$

$$\boxed{8 + i}$$

Conjugado

$$z = a + bi$$

Cambia el símbolo



solo a la parte

$$\overline{z} = a - bi$$

imaginaria

$$\overline{z+w} = \overline{z} + \overline{w}$$

$$\overline{3+2i} = 3 + \overline{2i} \\ = 3 - 2i$$

$$\frac{\overline{z}}{\overline{w}} = \frac{\overline{z}}{\overline{w}}$$

$$\overline{z \cdot w} = \overline{z} \cdot \overline{w}$$

$$\overline{\overline{z}} = z$$

Division

1)

$$\frac{z}{w} \cdot \frac{\overline{w}}{\overline{w}}$$

Siempre racionalizar
el denominador con
el signo contrario
ose el conjugado

(solo cambia el
signo de la parte
Imaginaria)

2)

$$\frac{2-i}{3+2i}$$

$$3+2i$$

Simbolo distinto

$$2-i \rightarrow 3-2i$$

$$3+2i \rightarrow 3-2i$$

$$\frac{(2-i)(3-2i)}{(3+2i)(3-2i)}$$

$$a \quad b \quad a \quad b$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$3^2 - (2i)^2$$

$$\frac{(2-i)(3-2i)}{9-4i^2}$$

$$9-4i^2$$

$$i^2 = -1$$

$$\frac{6-7i-3i+2i^2}{9+4}$$

$$9+4$$

$$\frac{6-7i-2}{13}$$

$$13$$

$$\frac{4-7i}{13}$$

$$13$$

$$\operatorname{Re}(z) = \frac{4}{13}$$

$$\operatorname{Im}(z) = \frac{-7}{13}$$

$$\frac{4}{13} - \frac{7}{13}i$$

1) Si z es cualquier número complejo, compruebe que $\frac{\overline{i+z}}{i-z} = -1$.

$$z = a+bi, \quad \overline{z} = 0+bi$$

$$\frac{i+\overline{z}}{i-z}$$

$$\overline{x+b} = \overline{x} + \overline{b}$$

$$\frac{\overline{i+z}}{i-z}$$

$$\overline{\overline{z}} = z$$

$$\overline{i} = -i$$

$$\frac{-i+z}{i-z}$$

$$\overline{-i} = i$$

$$\frac{-i+z}{-(\overline{-i+z})} = \boxed{-1}$$

2) Determine todos los números complejos x , expresados en su forma rectangular, que satisfacen:

$$a \cdot b = 0$$

$$a=0 \vee b=0 \quad (ix^2+x)\left(\frac{i \cdot x}{1-4i} - 1\right) = 0$$

Rectangular
Polar

$$ix^2+x=0$$

$$x(ix+1)=0$$

$$\boxed{x=0} \quad ix+1=0$$

$$ix=-1$$

$$x = \frac{-1}{i} \rightarrow \frac{-1 \cdot i}{i \cdot i} = \frac{-i}{-1} = i$$

$$i^2 = -1$$

$$= \frac{i}{1}$$

$$\boxed{x=i}$$

$$ix-1=0$$

$$ix=1$$

$$ix=1$$

$$ix=1$$

$$ix=1-i$$

$$x = \frac{1-i}{i} = \frac{(1-i) \cdot (-i)}{i \cdot (-i)} = \frac{-i+i^2}{-i^2} = \frac{-i-1}{1} = -1-i$$

$$S = \{0, i, -1-i\}$$

$$= \frac{-i+4i^2}{1}$$

$$-i^2 \rightarrow -1 \cdot -1 = 1$$

$$x = \frac{-i-4}{1} = \boxed{-4-i}$$

$$\rightarrow a + bi$$

5) Determine la forma rectangular de $z = \frac{-3+4i}{2-i} - i$.

$$\frac{-3+4i}{2-i} - i$$

$$\frac{-3+4i}{2-i} \cdot \frac{2+i}{2+i} - i$$

$(a-b)(a+b)$

$$\frac{(-3+4i)(2+i)}{2^2 - i^2} - i$$

$2^2 - i^2 \rightarrow 2 \cdot 2$

$$\frac{-6-3i+8i+4i^2}{4} - i$$

$4+1$

$$\frac{-6+5i-4}{5} - i$$

$$\frac{-10+5i}{5} - i$$

$$\frac{-2+i}{1} - i$$

~~5~~

$$-2+i-i$$

-2

$$z = -2 + 0i$$

3) Si se tiene que $w = \frac{2-5ai}{1+2i}$, determine todos los valores para el número real a , de tal forma que $\text{Im}(w) \neq 0$.

$$\frac{2-5ai}{1+2i} \quad \frac{1-2i}{1-2i}$$

$$\frac{(2-5ai)(1-2i)}{(1+2i)(1-2i)}$$

$$\frac{2-4i-5ai+10ai^2}{1^2-(2i)^2}$$

$$\frac{2-4i-5ai-10a}{1-4i^2}$$

$$\frac{(2-10a)+(-4-5a)i}{5}$$

$$\frac{2-10a}{5} + \frac{(-4-5a)}{5} i$$

$$\text{Im}(w) \neq 0$$

$$\frac{-4+5a}{5} \neq 0$$

$$-4+5a \neq 0$$

$$-5a \neq 4$$

$$a \neq -\frac{4}{5}$$

$$S = 1 - \left(-\frac{4}{5}\right)$$

4) Encuentre $x, y \in \mathbb{R}$ tales que $\frac{43+yi}{x-5i} = 4+3i$.

$$\frac{43+yi}{x-5i} = 4+3i$$

$$z = a+bi$$

$$a+bi = c+di$$

$$a=c \quad b=d$$

$$43+yi = (4+3i)(x-5i)$$

$$43+yi = 4x - 20i + 3ix - 15i^2$$

$$43+yi = 4x - 20i + 3ix + 15$$

$$43+yi = (4x+15) + (-20+3x)i$$

Re Im

Re

Im

$$43 = 4x + 15$$

$$4x = 28$$

$$x = 7$$

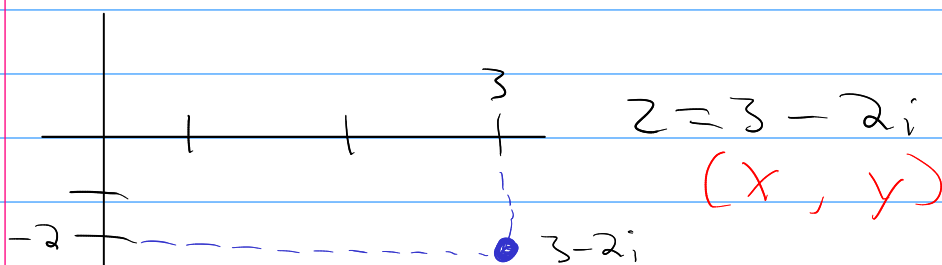
$$y = -20 + 3x$$

$$y = -20 + 3 \cdot 7$$

$$y = 1$$

$$\boxed{R/ \quad x=7 \quad \wedge \quad y=1}$$

Forma polar y exponencial



Rectangular $z = a+bi$

Polar $z = r \operatorname{cis}(\theta)$

$$|z| = r = \sqrt{a^2 + b^2} \quad \swarrow \quad \downarrow \quad \operatorname{Arg}(z) = \theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\downarrow \quad \cos(\theta) + i \sin(\theta)$$

$$1) = \overset{a}{3} - \overset{b}{2}i \quad \text{Polar} = r \text{cis}(\theta)$$

$$r = \sqrt{a^2 + b^2} \quad \theta = \tan^{-1}\left(\frac{a}{b}\right)$$

$$r = \sqrt{3^2 + (-2)^2} \quad \tan^{-1}\left(\frac{3}{-2}\right)$$

$$\approx 3,60 \quad \approx -33,69$$

$$z = 3,60 \cdot \text{cis}(-33,69)$$

$$z = \overset{a}{7} - \overset{b}{1}i$$

$$r = \sqrt{7^2 + (-1)^2} \quad \theta = \tan^{-1}\left(\frac{-1}{7}\right)$$

$$5\sqrt{2} \quad \approx -0,14$$

$$z = 5\sqrt{2} \cdot \text{cis}(-0,14)$$

$$5\sqrt{2} [\cos(-0,14) + i \sin(-0,14)]$$

$$5\sqrt{2} \cos(-0,14) + 5\sqrt{2} \cdot \sin(-0,14)i$$

De polar
a rectangular
solo hay
que expandir
cis y
resolver